Minku Kim

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Education

Oregon State University — *Corvallis, OR*
Ph.D. Candidate in Robotics — *Current*

University of Pennsylvania — *Philadelphia, PA*
M.S.E. in Mechanical Engineering and Applied Mechanics (4.0/4.0) — *May 2025*
- **Thesis**: Learning a Vision-Based Footstep Planner for Hierarchical Walking Control on Unstructured Terrain

Chung-Ang University — *Seoul, Korea*
B.S. in Mechanical Engineering with Honors — *Feb 2023*

Experience

Dynamic Robotics and Artificial intelligence Laboratory — *Corvallis, OR*
Graduate Research Assistant - Prof. Alan Fern — *Sep 2025 – Current*
- Developing loco-manipulation skills for humanoid robots using reinforcement learning and diffusion/flow-matching over contact, vision, and language inputs to enable robust, goal-conditioned whole-body interaction in complex environments

Agility — *Salem, OR*
Artificial Intelligence Intern — *Jun 2026 – Sep 2026*
- Developed and deployed a vision-based reinforcement learning policy for tote manipulation on Digit v4 hardware using reset distributions for faster learning and recovery.

Figueroa Lab, GRASP Lab — *Philadelphia, PA*
Graduate Research Assistant - Prof. Nadia Figueroa — *Mar 2025 – Current*
- Developed a real-time category-level 6D pose, shape, scale estimation, combining a flow-matching keypoint detector with Active Shape Models and a GPU-accelerated Matrix ADMM solver to achieve >100 Hz optimization for closed-loop manipulation
- Designed a real-time open-vocabulary multi-view transformer model integrating CLIP and DINOv3 for 6D pose estimation, achieving 13.7 FPS inference with an 8× lower memory usage than prior baselines such as PI3

Dynamic Autonomy and Intelligent Robotics Lab, GRASP Lab — *Philadelphia, PA*
Graduate Research Assistant - Prof. Michael Posa — *Jan 2024 – May 2025*
- Designed and deployed a vision-based hierarchical controller for the *Agility Robotics Cassie* bipedal robot, integrating a high-level RL footstep planner with a low-level operational space controller
- Built a full-stack RL pipeline in *Drake* for training, sampling, and hardware deployment, and benchmarked against a vision-based MPC footstep planner, demonstrating improved velocity tracking and success rates across diverse terrains

Publications

Humanoid Hanoi: Investigating Shared Whole-Body Control for Skill-Based Box Rearrangement — 2026
Under review
Minku Kim†, Kuan-Chia Chen†, Aayam Shrestha, Li Fuxin, Stefan Lee and Alan Fern

SAGE: Semantic And Geometric Estimation of 6D Object Pose from Multi-View Observations — 2026
Under review
Minku Kim†, Ho Jin Choi† and Nadia Figueroa

ASM-6D: Real-Time 6D Object Pose and Shape Estimation via Active Shape Models and ADMM — 2026
In Equivariant Systems Workshop at RSS 2025
Under review
Ho Jin Choi†, **Minku Kim**† and Nadia Figueroa

Learning a Vision-Based Footstep Planner for Hierarchical Walking Control — 2025
In IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids) [Oral Presentation]
Minku Kim, Brian Acosta, Pratik Chaudhari and Michael Posa

Honors and Awards

Oregon State University College of Engineering (COE) Scholarship — *2025*
Penn Engineering Outstanding Research Award — *2025*
CAU Winter Conference Da-Vinci Software Institute Excellence Award — *2022*
CAU Summer Conference Da-Vinci Software Institute Encouragement Award — *2021*

Technical Skills

Programming: Python, C/C++, MATLAB/Simulink, Git, Linux
Software/Frameworks: Pytorch, Tensorflow, OpenCV, ROS, LCM, Drake, MuJoCo, Isaac-Sim, Bazel, Docker, SLURM
Robotics: Legged Robot Control, Manipulation Control, Reinforcement Learning, Imitation Learning, Perception, Optimization